

HP Otto: Otto-D2

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VIEW IN BROWSER

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Summary

An R2D2 inspired body for the HP Otto Starter kit.

Hobby & Makers > RC & Robotics

Otto-D2

An R2D2 inspired body for the HP Otto Starter kit.

Otto-D2 uses all the components found in the HP Robots | Otto Starter - Creator Kit to build a complete Otto-based robot. It can also be used as an alternative body for the HP Robots | Otto Starter Santa Edition and the HP Robots | Otto Starter - Builder Kit.

Printing

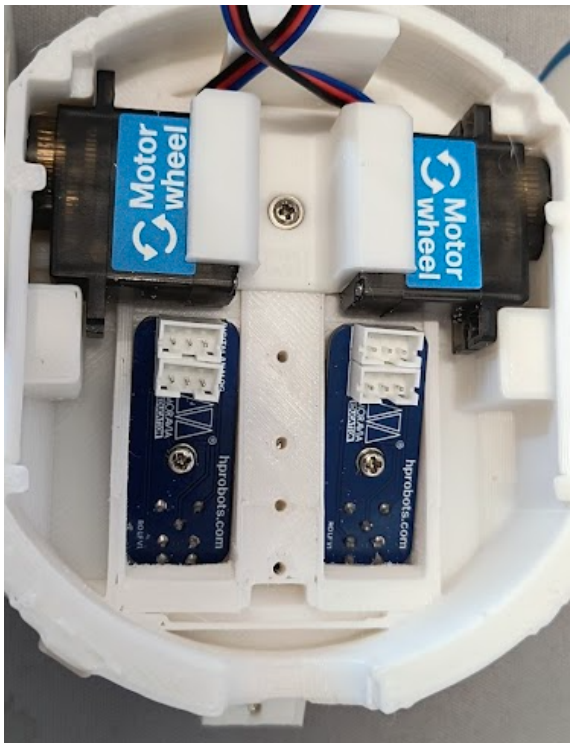
PLA, 15% infill, no supports. The middle section of the dome is transparent PLA to allow the LEDs to shine through. I used white for the body and silver for the rest of the dome. Various blue inserts give Otto-D2 a bit more colour.

Assembly

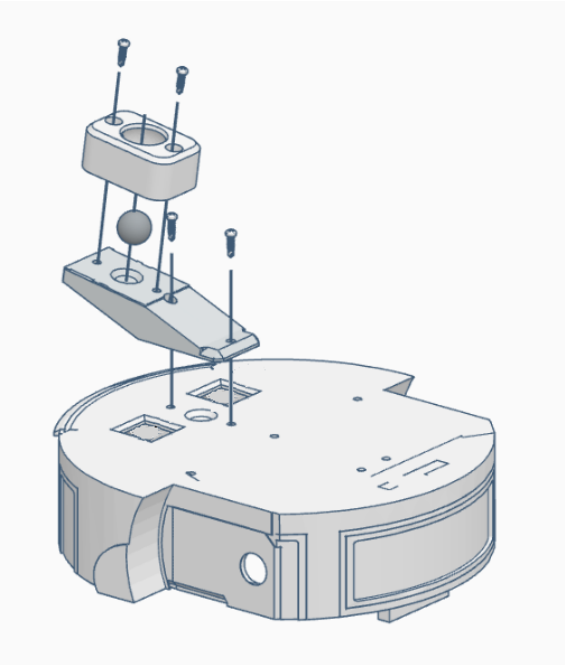
Note: Most of the assembly is very similar to the assembly detailed in the manual supplied with the starter kit and this manual should be referred to when assembling Otto-D2

1. Mount the line sensors in the base and secure with self tapping screws.
2. 

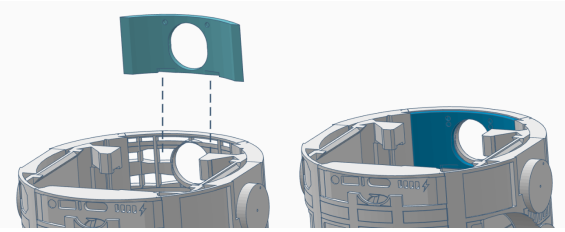
   Mount the servos and secure with the servo clamp, which is, in turn, secured with a self tapping screw.



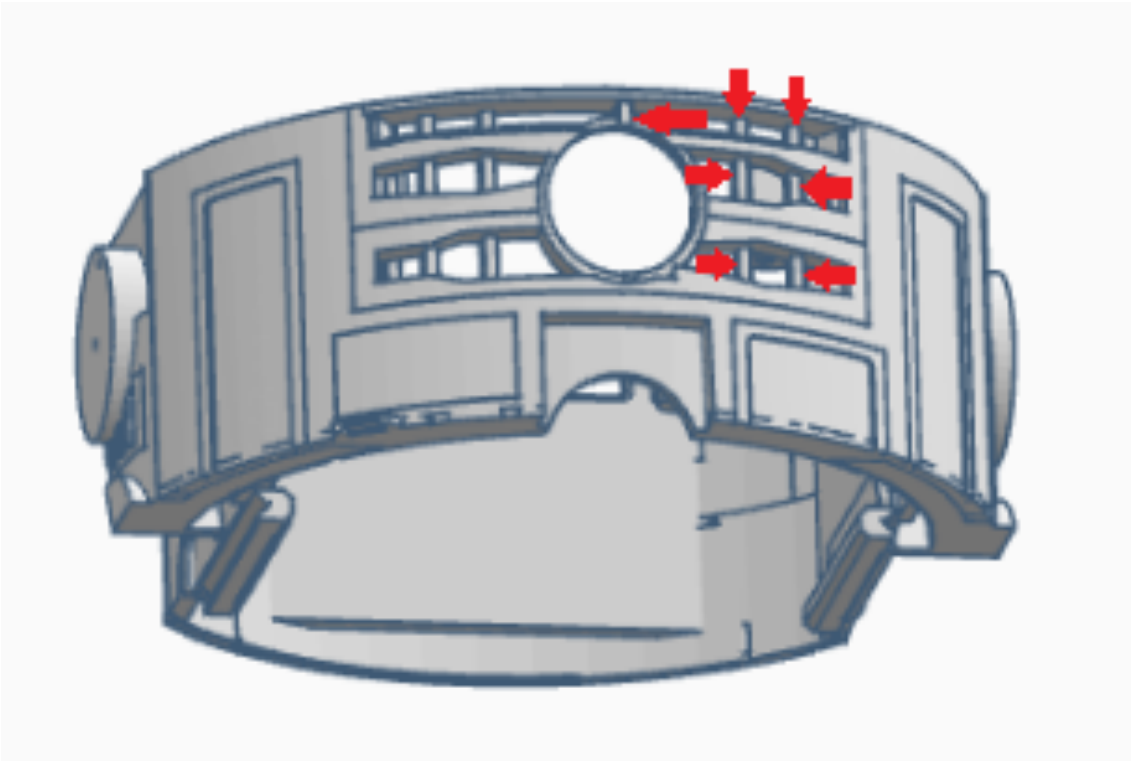
3. Secure the caster bracket and caster assembly to the underside of the base with more self tapping screws.



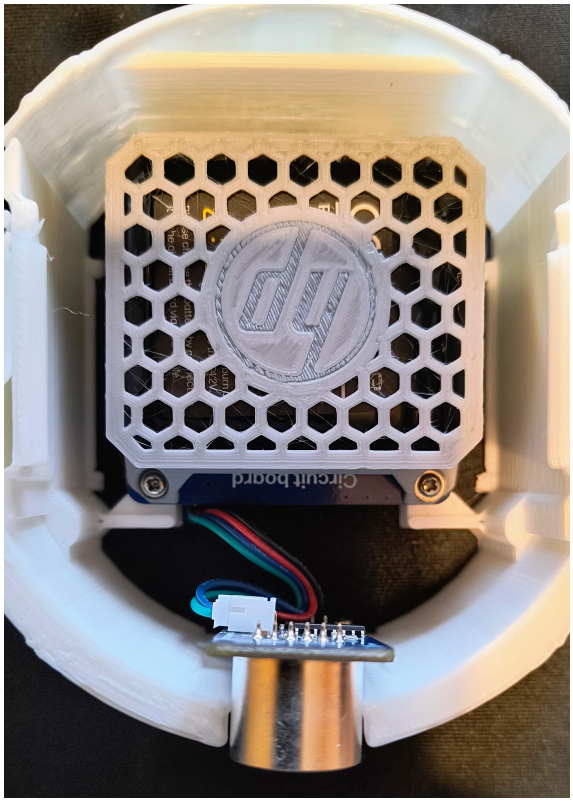
4. Slide the blue insert into the middle section.



5. The print supports indicated can be optionally nipped out after printing.

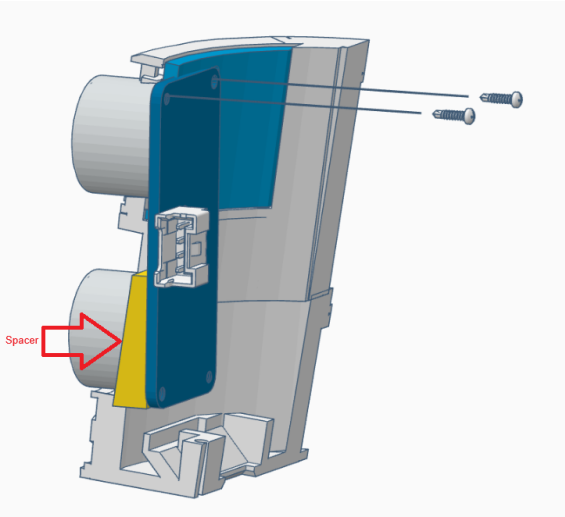


6. Assemble the 'inner module' following steps 8 to 10 of the HP Robots starter manual and secure the module inside the middle section. The ultrasonic sensor can be inserted in position at this time.



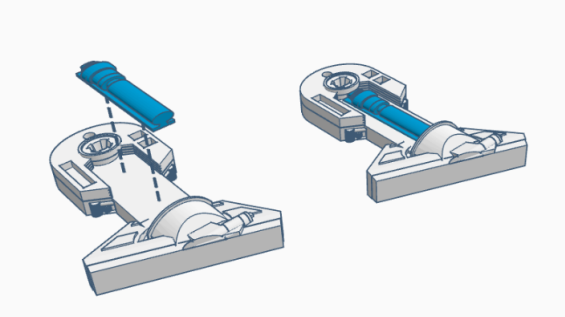
7. The base and middle can now be combined, by clipping them together. Make sure to feed all the wires from the base up the sides of the middle section so that they can be attached to the control

board. Use the spacer to ensure that the ultrasonic sensor is positioned at the correct angle.

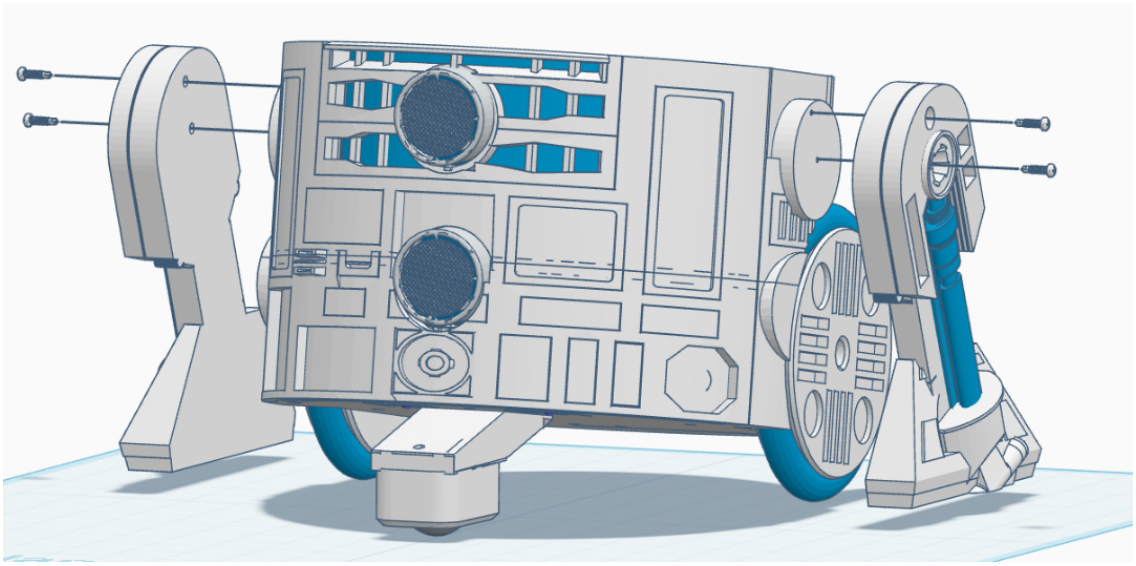


8. Attach the wheels, following steps 18 and 19 of the manual.

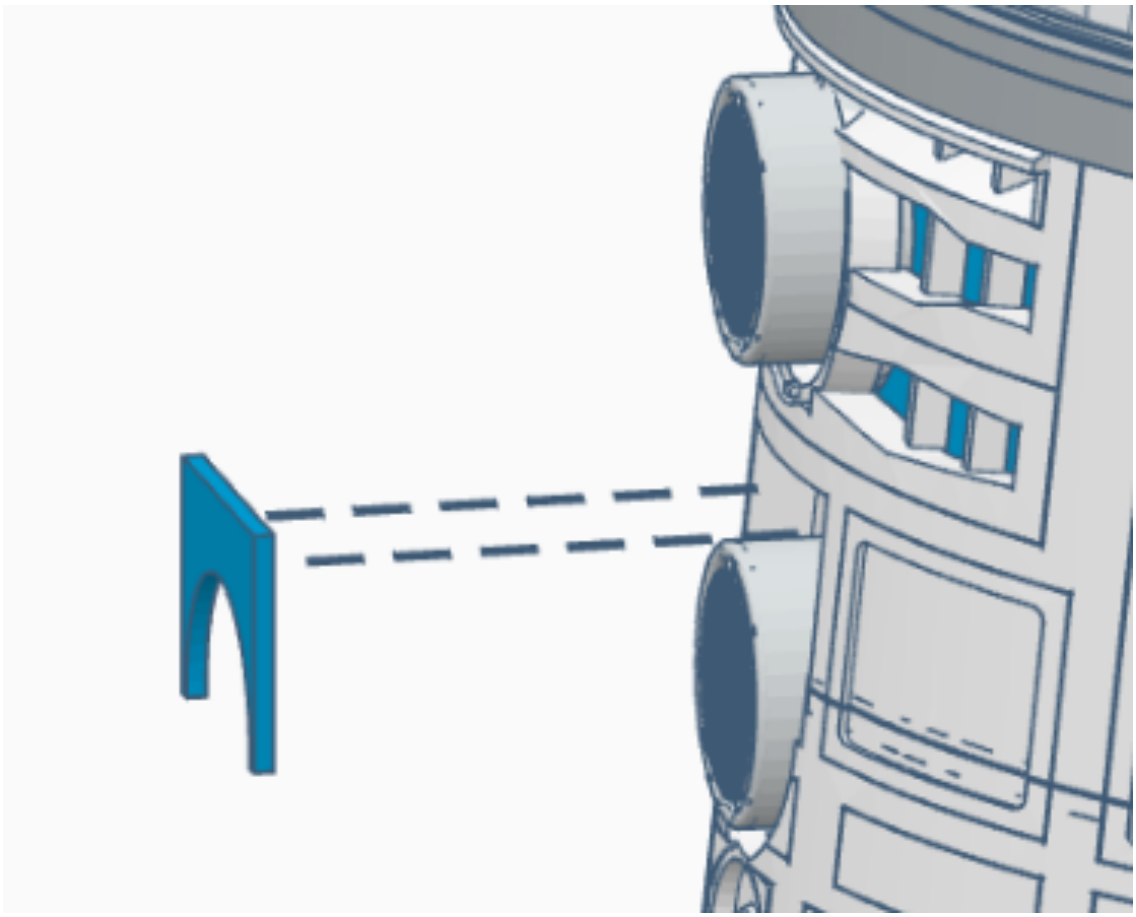
9. Assemble the legs by attaching the blue part with a drop of superglue.



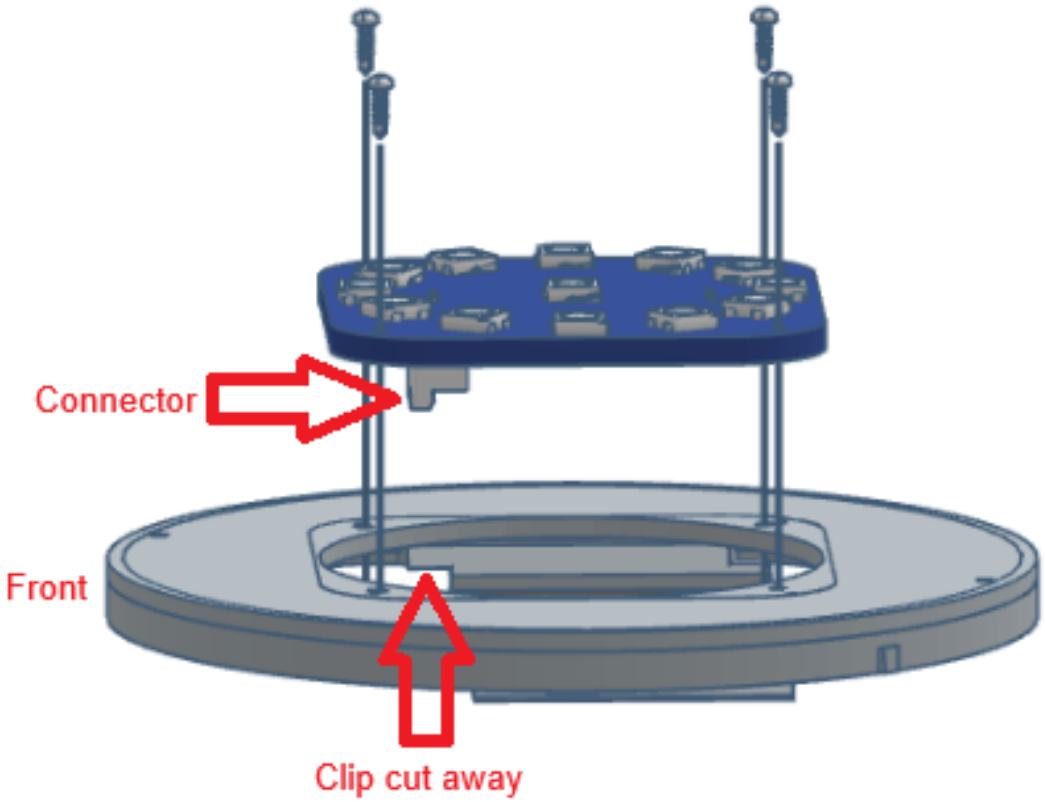
10. Now fit the legs, with 2 self tapping screws each.



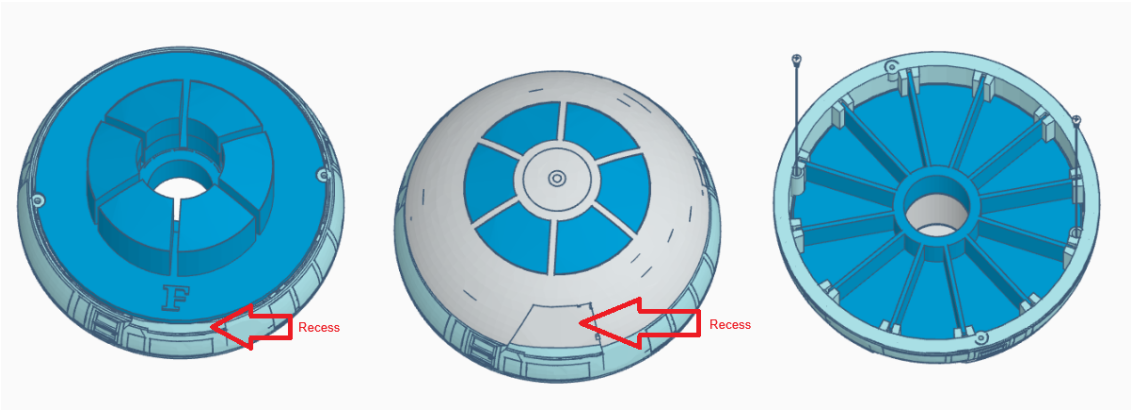
11. Finish off the body by attaching insert2 with a drop of superglue.



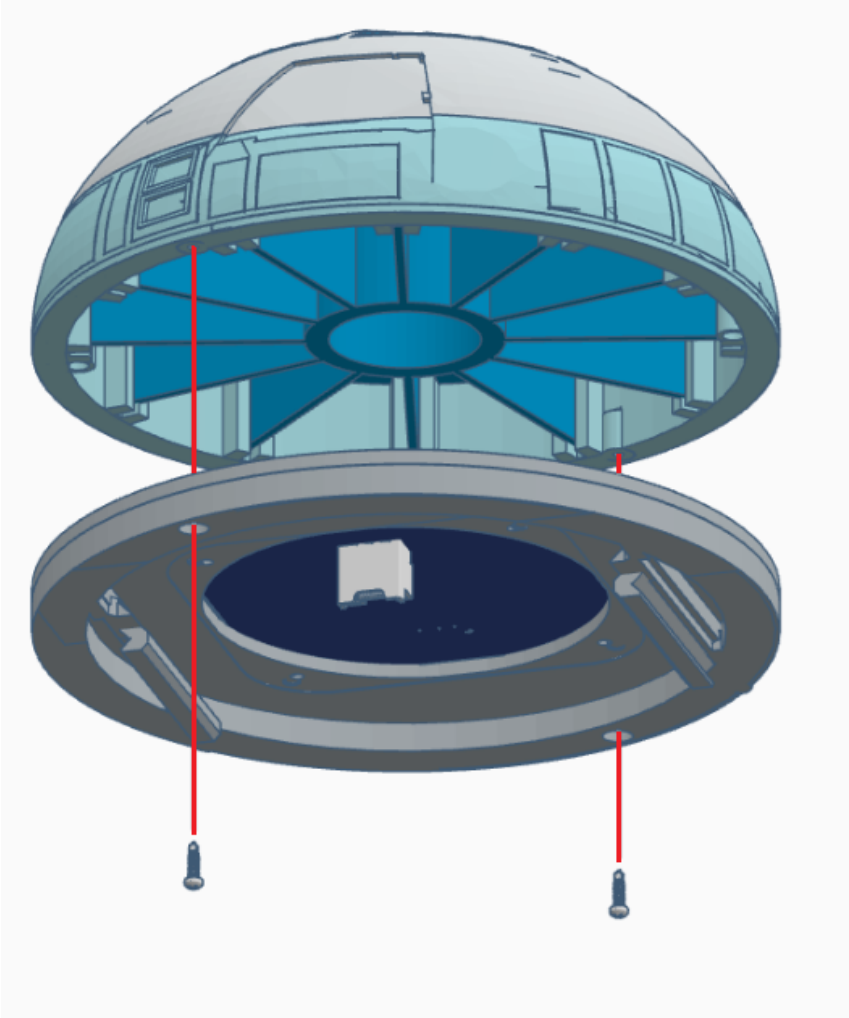
12. Now we come to assembly of the dome. This is made of 3 components which will be screwed together with, you guessed it, more self tapping screws. First the LED ring needs to be fitted. Take the dome base with the mounting clips facing down. Identify the front by looking for where the front section of the clip is cut away (see my helpful picture below). Attach the led ring on top of the dome base, with the connector facing down at the front.



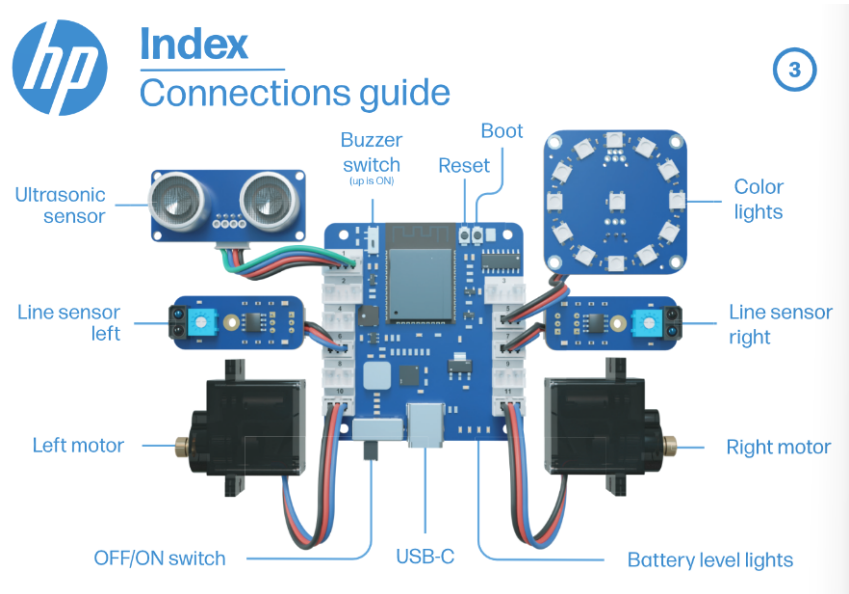
13. Now place the dome insert inside the dome middle, slide on the dome top and screw together.. (The insert has an F to indicate the front and the middle and top sections have a recessed part at the front, see diagram below)



14. Now attach this to the base with 2 more screws.

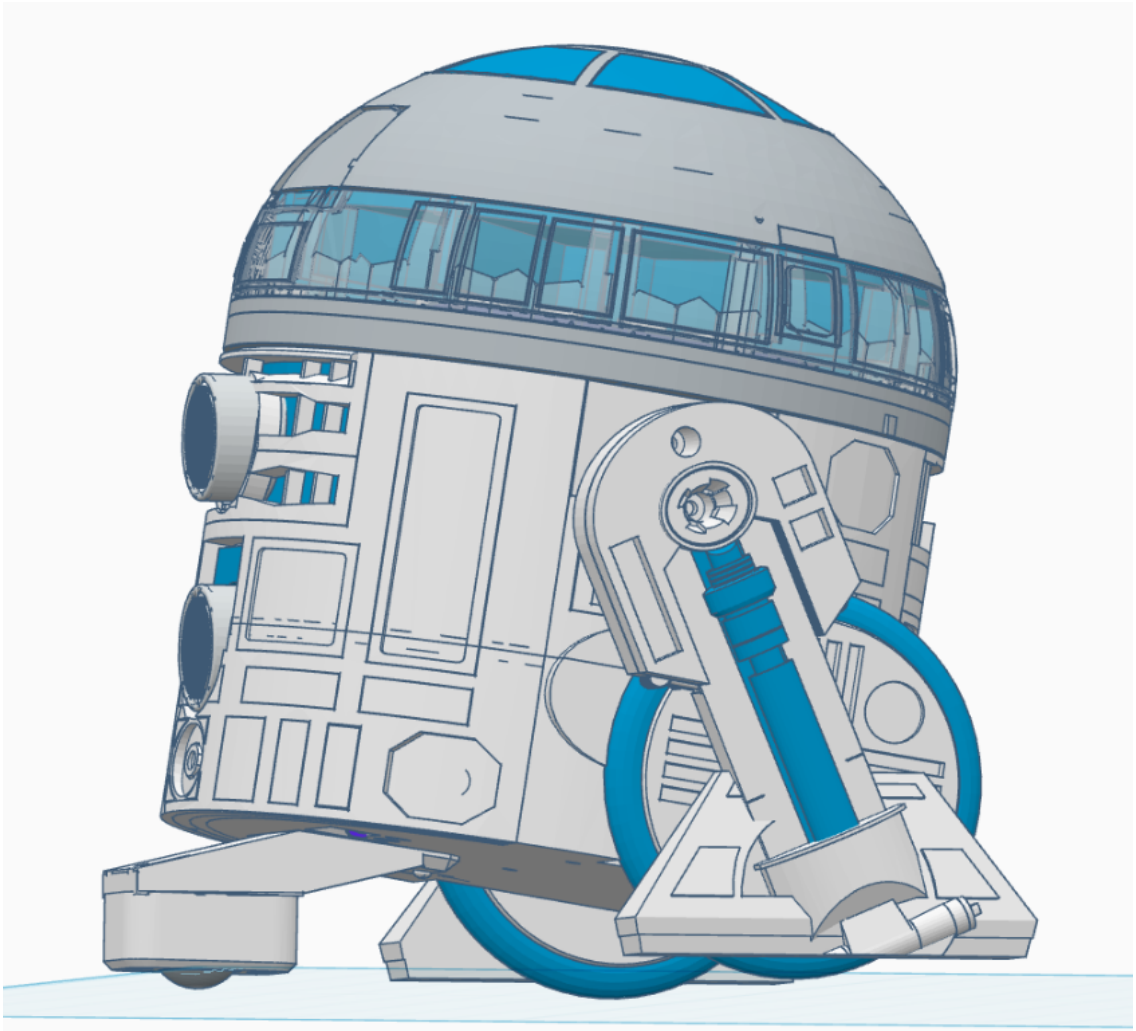


15. It's now time to make all the electrical connections. These can be found on page 3 of the manual.



(image used with permission)

16. Now clip the dome on to the body and you're done!



Otto-D2 can use exactly the same code as HP Otto starter. (https://hprobots.com/otto-robot/code/)

This remix is based on



HP Robots | Otto Starter

by HP Robots Otto



R2D2

by lubosh

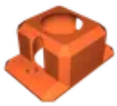
Model files



otto-d2-base.stl

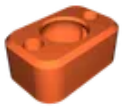


otto-d2-caster-bracket.stl



hp_starter_caster_s_r1.stl

☐ 2 choices from HP for the caster - take your pick!



hp_santa_caster_b_r2.stl

☐ 2 choices from HP for the caster - take your pick!



servo-clamp.stl

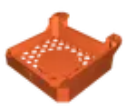
☐ (hp_starter_bracket_s_r1)



otto-d2-middle.stl



otto-d2-middle-blue-insert.stl



hp_starter_covermesh_b_r1.stl



otto-d2-wheels.stl

☐ Print 2



otto-d2-left-leg.stl



otto-d2-right-leg.stl



otto-d2-legs-blue-section.stl

☐ Print Two



otto-d2-ultrasonic-spacer.stl



otto-d2-middle-blue-insert-2.stl



otto-d2-dome-base.stl



otto-d2-dome-blue-insert.stl



otto-d2-dome-top.stl



otto-d2-dome-middle.stl

☐ transparent

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